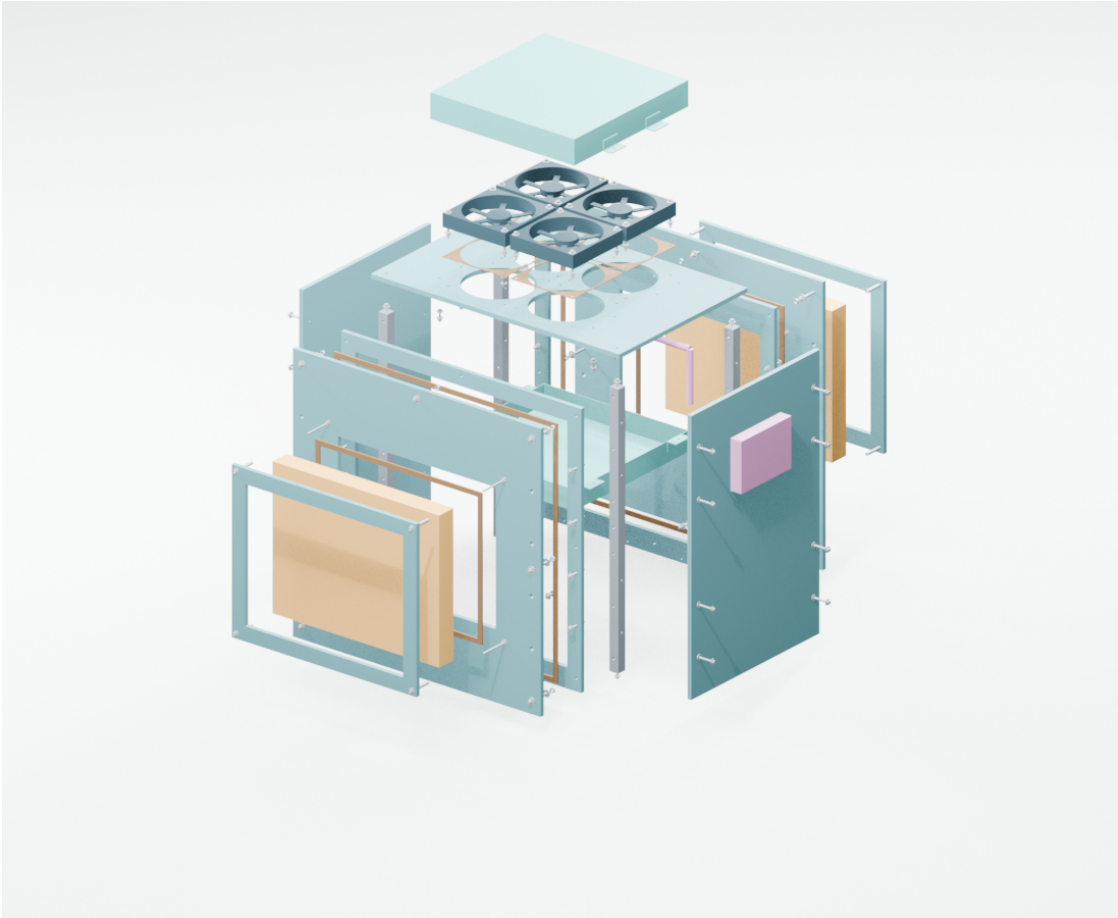


AQI Tower Engineering Handoff

Current D01 R03M integration with IH02 evidence



NOT FOR FABRICATION OR ENERGIZATION. Authoritative manufacturing geometry remains the FreeCAD R03M study; Blender and browser meshes are display derivatives.

The following review pages lead into the unchanged 41-page IH02 engineering reference. Return named decisions with evidence in TECHNICAL_REVIEW.xlsx. Do not mix legacy large-tower mains/HEPA assumptions with this 12 V indoor demonstrator.

Current pressure and geometry

Quantity	Current screen or evidence
Total flow	300 m3/h assumed, not measured
Parallel branches	4 fans at 75 m3/h each; 2 filters at 150 m3/h each
OEM fan head	27.63 Pa at stated curve condition
Nonfilter loss	13.31 Pa under K1 reducer scenario
Remaining filter head	14.32 Pa per parallel filter branch
CAD inventory	493 source objects; tessellated display derived from saved R03M
Display bounding envelope	595 x 504 x 631 mm including reservations and projecting hardware; not a fabrication specification

The two parallel filters do not add their pressure drops in series. Clean/loaded filter curves, installed/static fan curve compatibility, loss coefficients and branch equality remain unresolved. This screen is not an operating-point prediction.

Colour and animation interpretation

Teal identifies cabinet/guards, amber filters, dark blue fan envelopes, purple reserved control space. Motion shows part separation or hand-authored air paths, not velocity, turbulence, deposition or removal. Guard faces remain solid tessellations, not the proposed perforation geometry.

Review responsibilities and construction path

Discipline	Closure required
Mechanical	Guard strength/reach, seams and attachments; cabinet joints and stability; real filter tolerance, gasket compression, safe service access and whole mass/CG.
Electrical	Hazard allocation, independent protective functions, actual DC switching/isolation/protection, wire/connector ratings, cable restraints and enclosure. No guessed terminal assignments.
OEM-dependent	Exact fan/extension connector and clamping dimensions; delivered filter/frame properties. Sent enquiries are pending.
Commissioning and test	Confirm premises and targets; review assembly, isolation/restart and abnormal behaviour before collecting measured performance.

Critical path

Exact component evidence and accepted targets -> mechanical/electrical review -> revision-specific fabrication release -> supplier delivery and fabrication -> assembly and qualified commissioning -> airflow/pressure and controlled particle tests. These dependencies are not completion dates.

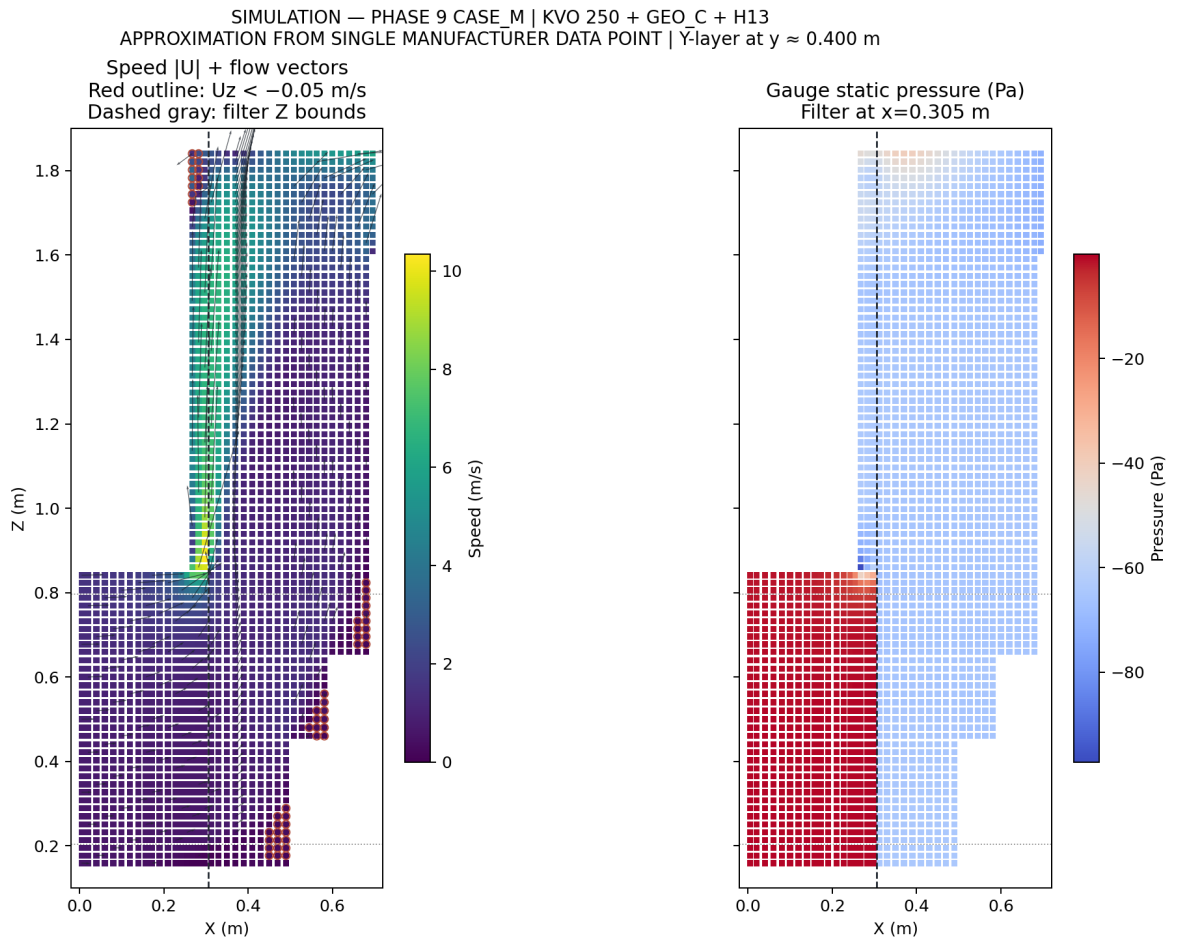
Outdoor boundary

Current fan temperature limits, enclosure and materials do not establish suitability for Indian outdoor heat or public misuse. Weather protection, tamper-resistant service access, anchoring and wind-dependent performance require a separate qualified design and approved site.

Files to use

Native CAD, STEP, 15 DXFs, current tools, measurement templates and source registers are in engineering/. TECHNICAL_REVIEW.xlsx combines parts, release, pressure and hardware sheets. The Blender file is for presentation only.

Historical CFD evidence



Existing Phase 9 CASE_M plot: KVO 250 + GEO_C + H13 approximation from a single manufacturer data point. Colours show calculated speed and gauge pressure for that historical case, not R03M performance.

GEO_C is an earlier air-path geometry trial. It is not the current manufacturing assembly. Historical CFD is partial/nonconverged; the later numerical bypass seal is not a proven physical seal. No new R03M CFD has been run for this delivery.

For the current prototype, use the pressure screen and obtain measured filter/installed-flow evidence. The following IH02 appendices are preserved unchanged.

Room duty and validation tools

Digital supplement. T01 remains OPEN: actual premises, accepted targets and instruments are not selected. The viewer calculates scenarios, not D01 CADR. First scope remains airborne particles; no gas, medical or outdoor coverage claims.

Mass balance and performance requirement

Volume $V = \text{area} \times \text{ceiling height}$. Ideal particle concentration ratio = $\exp(-\text{CADR} \times t / (60 \times V))$; time in minutes = $-60 \times V \times \ln(1-\text{reduction}) / \text{CADR}$. CADR is size-dependent effective particle-cleaned flow, not fan free-air flow or the 300 m³/h pressure-screen assumption. Source/influx, settling, ventilation and nonuniform mixing are excluded.

Illustrative room at 2.8 m height	90% in 30 min requires CADR	Ideal 90% time at assumed 150 m ³ /h
10 m ² / 28 m ³	128.9 m ³ /h	25.8 min
25 m ² / 70 m ³	322.4 m ³ /h	64.5 min
50 m ² / 140 m ³	644.7 m ³ /h	128.9 min

For a filtration-only mass balance, effective clean-air delivery cannot exceed actual installed airflow at 100% capture. The 25 m² example needs more than the assumed 300 m³/h screening flow. This is a conditional feasibility limit, not proof of delivered airflow. Accept room/target only after measured duty is known.

Tools carried forward

engineering/scripts/analysis/indoor_decay.py and its 8 synthetic tests are now included. This existing single-pair off/on fit estimates conditional additional particle-cleaned flow only if its assumptions are explicitly declared. It is not certified CADR; repeatability and uncertainty remain unevaluated. Preserve calibration, source/background, ventilation state, timing and measured room volume. Use the existing recorder schema, not the older PARTICLE_TIMESERIES_BLANK.csv interchange format.

A new room-scenario workbook tab and ROOM_MODEL_BASIS.md explain inputs, limitations and sources. The interactive person and room are visual references, not a modeled occupied ventilation field. No guard, electrical or structural approval has been granted by this supplement.

Primary references checked 5 October 2026: EPA Guide to Air Cleaners in the Home; CDC Appendix B Air. See ROOM_MODEL_BASIS.md for exact links and applicability limits.